

Credit - Drishti Ias

Mega Science Vision-2035 Report on Climate Research



Source: TH

Why in News?

The **Mega Science Vision-2035 (MSV) Report on Climate Research** raised an alarm that India has significantly lost its **ability to build indigenous scientific instruments**. This has made the country heavily **dependent on imported equipment for climate observations**, which is often run uncalibrated for years, severely compromising the quality of India's climate data.

What are the Key Highlights of the Report Climate Research?

- **Loss of Instrument-Making Culture:** India's climate science sector has almost entirely lost its **domestic instrument-manufacturing capacity**.
 - Consequently, critical environmental data relies on imported equipment.
 - Because imported instruments are routinely operated without proper calibration for years, it has led to **"incorrect data being reported in national and international journals."**
 - This is triggering global questions regarding the credibility of Indian scientific output.
- **Concerns over 'Uncontrolled' Renewables:** While India is racing toward its **Paris Agreement** commitment of **500 GW of non-fossil fuel capacity by 2030**, the report warns that the long-term climate consequences of massive solar and wind energy installations remain **"poorly understood" and require systematic, long-term environmental monitoring**.
- **Foreign Dependency for Earth System Models (ESMs):** An Earth System Model combines oceanic, land surface, and atmospheric data to simulate long-term climate trajectories.
 - Currently, Indian models are primarily adapted from the US and Europe. These foreign models are highly sensitive to **regional inputs that they were never designed to integrate**, making them inadequate for India's specific climatic conditions.
- **Procurement Challenges:** The mandatory use of the **Government e-Marketplace (GeM)** (prioritizes the lowest-bidding India-registered vendors) for procurement created challenges for scientific institutions, as vendors often failed to supply the customized, high-precision equipment required for advanced research.
 - While global tenders offered an alternative, they were frequently delayed by lengthy bureaucratic

procedures.

- Recognizing these concerns, the **Finance Ministry in June 2025 relaxed procurement norms**, allowing select institutions to procure scientific instruments outside GeM for purchases up to Rs 200 crore.
- **Other Identified Gaps:**
 - Inadequate research on **carbon capture and storage technologies** .
 - Weak integration of environmental surveillance data with **public health data** .
 - Absence of a comprehensive framework to embed climate considerations across broader public policy.

Other Structural Constraints of Indian Meteorological Forecasting

- **Geographical and Climatic Asymmetry** : Comparing India's forecasting accuracy directly with temperate nations like the US or UK is fundamentally flawed.
 - India's tropical climate is **inherently** more dynamic and unpredictable, lacking the systematic meteorological stability characteristic of higher latitudes.
- **The Climate Change Multiplier** : The ongoing climate crisis is **increasingly** fracturing historical weather patterns.
 - This systemic shift renders traditional predictive baseline models **significantly** less reliable, as future weather phenomena become highly erratic and divergent from past data.
- **The Macro-Micro Predictability Gap** : Forecasting accuracy is **heavily** contingent on geographical scale.
 - While the IMD can **relatively** accurately predict macro-level, cyclic phenomena (such as the broader monsoon trajectory or widespread heatwaves), anticipating sudden, hyper-local anomalies (like isolated cloudbursts) remains **exceptionally** difficult.

India's Meteorological Forecasting Initiatives

- **Mission Mausam** .
- **Bharat Forecasting System (BharatFS)** .
- **Impact-Based Forecasting (IBF)** .
- **AI-Enabled Forecast of Monsoon Advance** .
- **Doppler Weather Radars** .
- **SkyCast System**: Developed by the Ministry of Earth Sciences (MoES) under the Mission Mausam initiative, SkyCast is an advanced atmospheric remote sensing system designed for comprehensive aviation weather monitoring.
 - By unifying real-time measurements of fog, aerosols, turbulence, moisture, and visibility, the system aims to establish a weather-smart, "**fog-free**" **aviation framework** that maximizes passenger safety and drastically reduces flight delays, cancellations, and diversions during critical take-off and landing phases.

Mega Science Vision-2035 (MSV) Report

- **About:** It is a comprehensive roadmap prepared by the Indian climate research community.
- **Nodal Institution:** Indian Institute of Science (IISc), Bengaluru .
- **Submitted to:** The Office of the **Principal Scientific Adviser (PSA)** to the Union government.
- **Objective:** To outline the future trajectory, priorities, and "hopes and aspirations" of climate research in India, identifying critical gaps and funding requirements.

What are the Recommendations of the MSV Report?

- **Mega Projects:** The report proposes eight mega projects involving advanced observatories, dedicated satellites, field campaigns, in-situ monitoring networks, and indigenous sensor development.
 - These projects are planned in a phased manner up to 2035.
- **Indigenous Earth System Model:** India must build a domestic Earth System Model "**from first principles**," leveraging **Artificial Intelligence (AI) and Machine Learning** to suit context-specific **geographic and atmospheric conditions**.
- **Boost Domestic Manufacturing:** There must be targeted investments to revive the **domestic production of scientific instruments** and calibration technologies to reduce foreign dependence.
- **Expand Monitoring Networks:** Expand the country's surface weather stations, ocean buoy networks, and greenhouse gas monitoring system **s to eliminate current geographical "blind spots."**
- **Implement "Polluter Pays":** The report advocates devising scientific methods to estimate the **Social Cost of Carbon** (the economic cost of damages from an extra ton of CO₂) and implementing a strict **"polluter pays"** mechanism to prevent the atmosphere from being used as a dumping ground.

Conclusion

As India faces growing climate challenges such as heatwaves, erratic monsoons, and glacier melt, strengthening indigenous scientific capabilities is essential. Initiatives like the **Bharat Forecasting System (BharatFS)** and **AI-enabled Forecast of Monsoon Advance** underscore the need for self-reliance in climate science, while aligning procurement policies with research needs can enhance India's global leadership in weather and climate forecasting.

Drishti Mains Question:

"Scientific self-reliance is indispensable for climate resilience." Discuss.

Frequently Asked Questions (FAQs)

1. What is the Mega Science Vision-2035 (MSV) Report?

The MSV Report is a roadmap prepared by the Indian climate research community under the leadership of IISc Bengaluru and submitted to the Principal Scientific Adviser, outlining priorities and funding requirements for climate research up to 2035.

2. What is an Earth System Model (ESM)?

An Earth System Model integrates atmospheric, oceanic, land, and ecological processes to simulate climate behaviour and predict long-term climate change impacts.

3. What is the significance of the Polluter Pays Principle mentioned in the report?

The principle seeks to make polluters bear the economic cost of environmental damage by incorporating the Social Cost of Carbon into policy and regulatory frameworks.

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q “Momentum for Change: Climate Neutral Now” is an initiative launched by (2018)

- (a) The Intergovernmental Panel on Climate Change
- (b) The UNEP Secretariat
- (c) The UNFCCC Secretariat
- (d) The World Meteorological Organisation

Ans: (c)

Navachar Mantra Initiative



Source: PIB

Recently, the Union Minister of Skill Development and Entrepreneurship (MSDE) launched **Navachar Mantra** , a **flagship national programme** in New Delhi aimed at identifying, mentoring and scaling grassroots innovators.

- **About:** Launched by the **Ministry of Skill Development and Entrepreneurship (MSDE)**, Navachar Mantra is a national initiative to identify, nurture, and **amplify India's most promising grassroots** innovators and early-stage entrepreneurs.
- **Objective:** The programme creates pathways for innovators from **Tier-2 and Tier-3 towns, aspirational districts** , and **underserved geographies** to access mentorship, visibility, and growth opportunities.
- **Implementation:** It is executed by the **National Institute for Entrepreneurship and Small Business Development (NIESBUD)** in partnership with the **Foundation for Innovation and Technology Transfer (FITT)**, IIT Delhi .
- **Priority Sectors:** The initiative will support innovations in sectors such as **AgriTech, HealthTech, EdTech and Skilling, Climate and Sustainability, Rural Commerce and MSME Enablement** .
- **Support Framework:** Selected innovators will join a year-long programme with mentorship, webinars, showcases, investor interactions, and guidance on business development, market access, and scaling strategies.
- **Significance:** The initiative aims to build a strong pipeline of grassroots innovators, support local problem-solving, strengthen India's startup ecosystem and contribute to an **innovation-driven Viksit Bharat** .

Read More: [Skill Development for Future-Ready Competencies](#)

73rd NEC Plenary Meeting



Source: PIB

The 73rd Plenary Meeting of the **North Eastern Council (NEC)** outlined a comprehensive, multi-sectoral development strategy, emphasizing the **Ministry of Development of North-East Region (DoNER)** transition into a pivotal facilitator for regional growth aligned with the vision of Viksit

- **Sectoral Task Forces: Eight High-Level Task Forces** , constituted after the **72 nd NEC Plenary**, submitted action plans on **investment promotion, agriculture and horticulture, handicrafts, infrastructure and connectivity, economic corridors, livestock and fisheries, sports, and tourism** .
- **Implementation Mechanism:** A proposal has been initiated to establish a **High-Level Steering Committee** (comprising Union Ministers and relevant Chief Ministers) to strictly monitor the execution of the task force recommendations.
- **Financial Inclusion Milestones:** Grassroots financial penetration has rapidly accelerated with the integration of **6,500 previously unbanked villages** , consequently driving the region's **digital payment adoption via UPI to nearly 90%** .
- **Infrastructure & Tourism Corridors:** Development of iconic, high-yield geographical circuits is prioritized, specifically highlighting the **Matabari Tourism Circuit (Tripura)** and the **Sohra Tourism Circuit (Meghalaya)** to catalyze sustainable local livelihoods.
- **Soft Power & National Integration:** Soft diplomacy and scientific temper are being institutionalized through the **Ashtalakshmi Darshan Student Exchange Programme** and strategic **ISRO exposure visits** for regional youth.
- **One State One Product (OSOP):** Under the **OSOP initiative** , states are promoting unique products such as **Lakadong Turmeric (Meghalaya), Muga Silk (Assam), Kiwi (Arunachal Pradesh), Queen Pineapple (Tripura), and Organic Farming (Sikkim)** to strengthen exports and local incomes.
- **Governance Framework:** The Ministry of DoNER is operating under the **Three-C framework of Connect, Converge, and Catalyse** to accelerate regional policy implementation.
 - **DoNER's expenditure increased from around Rs 1,000 crore in 2022-23 to a budget allocation of Rs 6,500 crore for FY 2026-27** , reflecting greater focus on **North East development**.
 - The **Rising North East Investment Summit** generated investment commitments worth **Rs 4.5 lakh crore** .

North Eastern Council (NEC)

- The **NEC** is a **statutory body** established under the **North Eastern Council Act, 1971** , and functions under the administrative control of the **MDoNER** .
- It serves as the nodal agency for the **economic and social development of the North Eastern Region** , comprising **Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, and Tripura** .
- The Council consists of the Governors and Chief Ministers of all eight member states along with **three members nominated by the President of India** .
- The **Union Home Minister serves as the ex-officio Chairman** , while the **Minister of State (Independent Charge), DoNER acts as the ex-officio Vice-Chairman** of the Council.

Read more: [Bringing India's Northeast Region to Mainstream](#)

India-UK Critical Minerals Global Supply Chain Observatory



Source: PIB

India and the UK launched the **India-UK Critical Minerals Global Supply Chain Observatory (GSCO)** in New Delhi to strengthen resilient and diversified supply chains for **critical minerals** essential for clean energy technologies and advanced manufacturing.

- **Collaborative Architecture:** The GSCO is a tripartite joint initiative spearheaded by **TEXMiN** (Technology Innovation in Exploration & Mining Foundation at IIT-ISM Dhanbad), the **Indian Institute of Technology (ISM) Dhanbad** , and the **University of Cambridge** .
 - The initiative stems from the **India-UK Technology Security Initiative** . GSCO was announced during the **India-UK Prime Ministers' bilateral engagement in October 2025** and subsequently formalised through a **Research Collaboration Agreement signed in March 2026**.
- **Core Operational Mandate:** The GSCO functions as a **data-driven digital platform dedicated to monitoring global critical mineral flows** , mapping supply risks, anticipating market disruptions, and providing strategic intelligence to policymakers, industry leaders, and researchers.
- **Strategic Significance for India:**
 - **Supports the National Critical Mineral Mission (NCMM)** by providing data for mineral exploration, processing, and recycling initiatives.
 - **Reduces supply chain vulnerabilities** by helping India monitor and mitigate risks arising from the concentration of critical mineral processing in a few countries, particularly China.
 - **Facilitates the clean energy transition** by ensuring secure access to key minerals such as **lithium, cobalt, nickel, and rare earth elements** , essential for EVs, battery storage, and renewable energy technologies.
- **Broader India-UK Relations:**
 - The **GSCO supports the India-UK Vision 2035** roadmap by strengthening cooperation in economic growth, technology, and resource security.
 - It complements the **India-UK Comprehensive Economic and Trade Agreement (CETA)** , with a focus on resilient and secure supply chains.

Read more: [India-UK Vision 2035 and CETA](#)

India Signs BrahMos Missile Deal with Vietnam



Source: TH

Recently, the Defence Secretary said at the **Shangri-La Dialogue** in Singapore that India has signed a **BrahMos missile deal with Vietnam** , while a similar agreement with **Indonesia** is in the final stages following a March 2026 agreement.

- **Philippines as First Buyer:** The Philippines was the first foreign buyer, signing a contract worth nearly **\$375 million in 2022**, highlighting India's export of the missile system in Southeast Asia.
- **Technology Sharing Policy:** India shares advanced defence technologies with countries it regards as friendly partners, including **ASEAN nations**, emphasizing trust and strategic alignment.
- **Regional Significance:** Several ASEAN members, including the Philippines and Vietnam, have overlapping maritime claims with China in the **South China Sea** , making BrahMos exports significant for regional defence partnerships.
- **Shangri-La Dialogue:** It is an important annual inter-governmental security forum held in **Singapore** . It brings together defence ministers, military officials and strategic experts to discuss **Indo-Pacific** and global security issues.

BrahMos Missile

- **About:** The BrahMos missile is a **supersonic cruise missile** jointly developed by India and Russia, integrating Indian guidance and control systems with Russian propulsion technology for high performance and reliability. Its name is derived from the **Brahmaputra** and **Moskva** rivers.
- **Historical Milestones:** BrahMos was inducted into the **Navy in 2005** , the **Indian Army in 2007** , and first successfully launched from an **IAF Sukhoi-30 MKI fighter in 2017** .
- **Type & Versatility:** It is **land, air, sea, and submarine-launched** , with multiple versions featuring extended ranges and evolving onboard sensing capabilities, making it versatile across operational domains.
- **Speed:** The missile travels at **supersonic speeds of around Mach 2.8-3** , enabling rapid engagement of targets.
- **Significance:** BrahMos' high-speed flight stability, advanced guidance, and terminal accuracy demonstrate India's progress in **indigenous missile technology** under the **Atmanirbhar Bharat initiative** .

Read More: [BrahMos Missiles](#)